

# 22AIE111 OBJECT ORIENTED PROGRAMMING IN JAVA

## S2 B.Tech CSE-AI

### Department of Computer Science and Engineering

### Amrita School of Computing, Amritapuri Campus

### CASE STUDY

### PHASE – I (Analysis)

### Problem Identification & Class Mapping (CO1)

Group No : 12

Team Members :

| Roll No          | Student Name    |
|------------------|-----------------|
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1. Problem Title : **Hostel Management System**

2. Problem Description :

Manual hostel allocation processes lead to record mismatches, allocation conflicts, and delays in accommodation management. There is no centralized digital system to handle student applications, room assignment, occupancy tracking, and record maintenance. This system aims to digitize the entire process, ensuring transparency, efficiency, and accuracy.

3. Identified Entities (Classes)

| Class Name | Description                                                                      |
|------------|----------------------------------------------------------------------------------|
| Student    | Represents a student who requests hostel accommodation.                          |
| Warden     | Represents the warden who allocates rooms and oversees hostel operations.        |
| Admin      | Manages hostels, blocks, and overall system configuration.                       |
| Hostel     | Represents a hostel building with basic info like name, total capacity, etc.     |
| Room       | Represents a room within a hostel (room number, type, availability).             |
| Allocation | Manages the mapping between students and allocated rooms.                        |
| System     | Maintains all records and handles core logic (application, allocation, updates). |

4. Class Attributes and Behaviours

| Class Name | Attributes                                                                | Methods(behaviours)                                                                     |
|------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Student    | (-)Name<br>(-)Roll number<br>(-)Applied                                   | (+)applyForRoom()<br>(+)checkStatus()                                                   |
| Warden     | (-)Wname<br>(-)wardenId<br>(-)hostelName                                  | (+)allotRoom(Student s, Room r)<br>(+)viewEmptyRooms()<br>(+)cancelAllotment(Student s) |
| Admin      | (-)name<br>(-)adminId<br>(-)password                                      | (+)addHostel(Hostel h)<br>(+)removeHostel(Hostel h)<br>(+)viewAllStudents()             |
| Hostel     | (-)hostelName<br>(-)totalRooms<br>(-)address                              | (+)getFreeRoomsCount()<br>(+)showHostelDetails()                                        |
| Room       | (-)roomNumber<br>(-)hostelName<br>(-)isOccupied                           | (+)bookRoom()<br>(+)vacateRoom()<br>(+)checkAvailability()                              |
| Allocation | (-)studentRoll<br>(-)roomNumber                                           | (+)getStdRollNumber()<br>(+)getRoomNumber()                                             |
| System     | (-)studentsList<br>(-)wardensList<br>(-)hostelsList<br>(-)allocationsList | (+)registerStudent(Student s)<br>(+)processApplication(Student s)<br>(+)autoAllocate()  |